

- 29** One wave has an amplitude of $2A$. A second wave has an amplitude of $\frac{A}{2}$. The waves are otherwise identical.

The two waves travel in opposite directions and overlap.

What is the ratio $\frac{\text{maximum amplitude of combined wave}}{\text{minimum amplitude of combined wave}}$?

A $\frac{3}{2}$

B $\frac{5}{3}$

C $\frac{5}{2}$

D 4

- 30** Light of wavelength 680 nm is incident normally on a diffraction grating with 450 lines mm^{-1}